SHARAVANAN R

B.E. Electronics & Communication Engineering ● Rajalakshmi Institute of Technology ● Chennai, India
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OBJECTIVE

ECE undergraduate and systems builder specializing in **Robotics**, **Embedded Systems**, and **Backend architectures**. Focused on engineering high-performance, cost-optimized solutions by integrating precision bio-signal acquisition, edge computing, and real-time IoT control logic to solve complex physical problems.

EDUCATION

B.E. in Electronics and Communication Engineering — **Aug 2024 – Expected May 2028**
Rajalakshmi Institute of Technology, Chennai — CGPA: **8.4**

Higher Secondary Certificate (HSC) — **Completed May 2024**
Jayapriya Vidyalaya Matric Higher Secondary School, Gopalapuram — Score: **91.33%**

Secondary School Leaving Certificate (SSLC) — **Completed May 2022**
Jayapriya Vidyalaya Matric Higher Secondary School, Gopalapuram — Score: **90.8%**

PROFESSIONAL EXPERIENCE

1M1B Green Skills Academy (AICTE & Salesforce) — **May 2025 – June 2025**
Green Intern — Sustainability & Digital Technology — Remote

- Completed a 60-hour intensive program, executing a 20-hour Live Project focused on technology-driven sustainability initiatives.
- Leveraged **Tableau** and Salesforce ecosystems to analyze environmental datasets and visualize data-driven metrics.

TECHNICAL SKILLS

- **Languages & OS:** Python, C, C++, Dart (Flutter), Java, Linux/Unix, ROS2.
- **Hardware & Embedded:** ESP32, Arduino Nano, AD620 Instrumentation Amp, ADS1115 16-bit ADC, Peltier Modules, I^2C, MQTT.
- **Software, Backend & AI:** Firebase Firestore (NoSQL), Google Gemini API, Tableau, Streamlit, Git, Signal Processing.

PROJECTS

NeuroNode — *AI/ML & Embedded Systems Engineer (2nd Prize, Innovista RIT)*

- **Led a research team** to develop real-time emotion classification capabilities, reducing BCI hardware costs by 96% (INR 25k to INR 900).
- Enabled the system to autonomously process EMG signals via AD620/ADS1115 and Gemini API, achieving <2s latency.

INNSOLE — *Embedded Systems Engineer*

- Developed capabilities for precision thermal regulation in off-grid environments to maintain critical temperatures for medication storage.
- Designed a modular CAD/3D-printed chassis utilizing Peltier modules and DC-DC buck regulation to maximize thermal isolation and battery longevity.

KERA AIR — *Full-Stack IoT Engineer (Winner, INR 6,000 Hackathon Prize)*

- Developed autonomous air filtration capabilities, enabling the system to trigger actuator controls based on real-time MQ-series sensor data.
- **Led the integration** of MQTT data brokering and a Flutter UI for remote system override and telemetry.

Print-E — *Backend & IoT Engineer*

- Led the backend architecture of a wireless printing ecosystem, integrating ESP32 firmware with Firebase Firestore (NoSQL) to automate job queuing and eliminate manual dispatch.

PUBLICATIONS & PATENTS

- **Conference Presentation:** Presented NeuroNode research at the AICTE International Technical Conference, Rajalakshmi Institute of Technology (2025).